

EAS 501 Lecture Notes 09/21/2026

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Outline

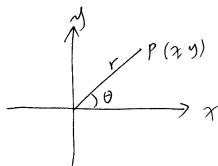
- Chapter 16. Scalar and Vector Field Theory

16.7 Non Cartesian coordinate systems

Polar coordinate (2-D):
$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \end{cases}$$

Cylindrical coordinate (3-D):
$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \\ z = z, \end{cases}$$

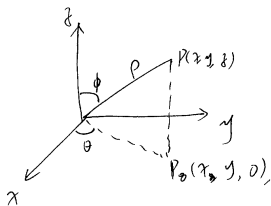
Spheric coordinate (3-D):
$$\begin{cases} x = \rho \sin \phi \cos \theta, \\ y = \rho \sin \phi \sin \theta, \\ z = \rho \cos \phi, \end{cases}$$



polar coordinate

$$x = r \cos \theta,$$

$$y = r \sin \theta,$$



Spheric coordinate

$$x = \rho \sin \phi \cos \theta,$$

$$y = \rho \sin \phi \sin \theta,$$

$$z = \rho \cos \phi,$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$ in the polar coordinate

Consider $u = f(x, y) = g(r, \theta)$, with $x = r \cos \theta$, $y = r \sin \theta$.

We see that $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1} \frac{y}{x}$, so that

$$r_x = \frac{x}{\sqrt{x^2 + y^2}} = \cos \theta, \quad r_y = \frac{y}{\sqrt{x^2 + y^2}} = \sin \theta, \quad (1)$$

$$\theta_x = \frac{-\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = \frac{-y}{x^2 + y^2} = -\frac{\sin \theta}{r}, \quad (2)$$

$$\theta_y = \frac{\frac{1}{x}}{1 + \frac{y^2}{x^2}} = \frac{x}{x^2 + y^2} = \frac{\cos \theta}{r}. \quad (3)$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

Therefore, we are able to derive

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} \\ &= \frac{\partial u}{\partial r} \cdot \cos \theta + \frac{\partial u}{\partial \theta} \cdot \left(\frac{-\sin \theta}{r} \right),\end{aligned}\tag{4}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial y} \\ &= \frac{\partial u}{\partial r} \cdot \sin \theta + \frac{\partial u}{\partial \theta} \cdot \left(\frac{\cos \theta}{r} \right).\end{aligned}\tag{5}$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

Moreover, we get

$$\begin{aligned}
 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial r \partial x} \cdot \cos \theta + \frac{\partial u}{\partial r} \cdot \frac{\partial \cos \theta}{\partial x} \\
 &\quad + \frac{\partial^2 u}{\partial \theta \partial x} \cdot \left(\frac{-\sin \theta}{r} \right) + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \left(\frac{-\sin \theta}{r} \right)}{\partial x} \\
 &= \left(\frac{\partial^2 u}{\partial^2 r} \cdot \frac{\partial r}{\partial x} + \frac{\partial^2 u}{\partial r \partial \theta} \cdot \frac{\partial \theta}{\partial x} \right) \cdot \cos \theta \\
 &\quad + \frac{\partial u}{\partial r} \cdot \left(\frac{\partial \cos \theta}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial \cos \theta}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} \right) \\
 &\quad + \left(\frac{\partial^2 u}{\partial r \partial \theta} \cdot \frac{\partial r}{\partial x} + \frac{\partial^2 u}{\partial^2 \theta} \cdot \frac{\partial \theta}{\partial x} \right) \cdot \left(\frac{-\sin \theta}{r} \right) \\
 &\quad + \frac{\partial u}{\partial \theta} \cdot \left(\frac{\partial \left(\frac{-\sin \theta}{r} \right)}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial \left(\frac{-\sin \theta}{r} \right)}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} \right), \quad (6)
 \end{aligned}$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

$$\begin{aligned}
\frac{\partial^2 u}{\partial x^2} &= \left(\frac{\partial^2 u}{\partial^2 r} \cdot \cos \theta + \frac{\partial^2 u}{\partial r \partial \theta} \cdot \left(\frac{-\sin \theta}{r} \right) \right) \cdot \cos \theta \\
&\quad + \frac{\partial u}{\partial r} \cdot \left(0 \cdot \cos \theta + \frac{\partial \cos \theta}{\partial \theta} \cdot \left(\frac{-\sin \theta}{r} \right) \right) \\
&\quad + \left(\frac{\partial^2 u}{\partial r \partial \theta} \cdot \cos \theta + \frac{\partial^2 u}{\partial^2 \theta} \cdot \left(\frac{-\sin \theta}{r} \right) \right) \cdot \left(\frac{-\sin \theta}{r} \right) \\
&\quad + \frac{\partial u}{\partial \theta} \cdot \left(\frac{\partial \left(\frac{-\sin \theta}{r} \right)}{\partial r} \cdot \cos \theta + \frac{\partial \left(\frac{-\sin \theta}{r} \right)}{\partial \theta} \cdot \left(\frac{-\sin \theta}{r} \right) \right) \\
&= \frac{\partial^2 u}{\partial^2 r} \cdot \cos^2 \theta - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
&\quad + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial^2 \theta} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta}. \tag{7}
\end{aligned}$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

Similarly, we have

$$\begin{aligned}
 \frac{\partial^2 u}{\partial y^2} &= \frac{\partial^2 u}{\partial r \partial y} \cdot \sin \theta + \frac{\partial u}{\partial r} \cdot \frac{\partial \sin \theta}{\partial y} \\
 &\quad + \frac{\partial^2 u}{\partial \theta \partial y} \cdot \left(\frac{\cos \theta}{r}\right) + \frac{\partial u}{\partial \theta} \cdot \frac{\partial(\frac{\cos \theta}{r})}{\partial y} \\
 &= \left(\frac{\partial^2 u}{\partial^2 r} \cdot \frac{\partial r}{\partial y} + \frac{\partial^2 u}{\partial r \partial \theta} \cdot \frac{\partial \theta}{\partial y}\right) \cdot \sin \theta \\
 &\quad + \frac{\partial u}{\partial r} \cdot \left(\frac{\partial \sin \theta}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial \sin \theta}{\partial \theta} \cdot \frac{\partial \theta}{\partial y}\right) \\
 &\quad + \left(\frac{\partial^2 u}{\partial r \partial \theta} \cdot \frac{\partial r}{\partial y} + \frac{\partial^2 u}{\partial^2 \theta} \cdot \frac{\partial \theta}{\partial y}\right) \cdot \left(\frac{\cos \theta}{r}\right) \\
 &\quad + \frac{\partial u}{\partial \theta} \cdot \left(\frac{\partial(\frac{\cos \theta}{r})}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial(\frac{\cos \theta}{r})}{\partial \theta} \cdot \frac{\partial \theta}{\partial y}\right), \tag{8}
 \end{aligned}$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

$$\begin{aligned}
\frac{\partial^2 u}{\partial y^2} &= \left(\frac{\partial^2 u}{\partial^2 r} \cdot \sin \theta + \frac{\partial^2 u}{\partial r \partial \theta} \cdot \left(\frac{\cos \theta}{r} \right) \right) \cdot \sin \theta \\
&\quad + \frac{\partial u}{\partial r} \cdot \left(0 \cdot \sin \theta + \cos \theta \cdot \left(\frac{\cos \theta}{r} \right) \right) \\
&\quad + \left(\frac{\partial^2 u}{\partial r \partial \theta} \cdot \sin \theta + \frac{\partial^2 u}{\partial^2 \theta} \cdot \left(\frac{\cos \theta}{r} \right) \right) \cdot \left(\frac{\cos \theta}{r} \right) \\
&\quad + \frac{\partial u}{\partial \theta} \cdot \left(\frac{-\cos \theta}{r^2} \right) \cdot \sin \theta + \frac{-\sin \theta}{r} \cdot \frac{\cos \theta}{r} \\
&= \frac{\partial^2 u}{\partial^2 r} \cdot \sin^2 \theta + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
&\quad + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial^2 \theta} + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta}. \quad (9)
\end{aligned}$$

Transformation of $\Delta = \partial_{xx} + \partial_{yy}$, continued

A combination of (7) and (9) leads to

$$\begin{aligned}
 \Delta u &= \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial y^2} \\
 &= \frac{\partial^2 u}{\partial^2 r} \cdot \cos^2 \theta - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
 &\quad + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial^2 \theta} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} \\
 &\quad + \frac{\partial^2 u}{\partial^2 r} \cdot \sin^2 \theta + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
 &\quad + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial^2 \theta} + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} \\
 &= \frac{\partial^2 u}{\partial^2 r} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial^2 \theta}.
 \end{aligned} \tag{10}$$

Transformation in the cylindrical coordinate

Similarly, in the 3-D cylindrical coordinate, we have

$$\begin{aligned}\Delta u &= \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \\ &= \frac{\partial^2 u}{\partial^2 r} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial^2 \theta} + \frac{\partial^2 u}{\partial z^2}.\end{aligned}\tag{11}$$

Transformation in the spherical coordinate

In the 3-D spherical coordinate, we have (**exercise**)

$$\begin{aligned}\Delta u &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \\ &= \frac{1}{\rho^2} \left(\frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial u}{\partial \rho} \right) + \frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial u}{\partial \phi} \right) + \frac{1}{\sin^2 \phi} \frac{\partial^2 u}{\partial \theta^2} \right) \quad (12)\end{aligned}$$